

Contestant Number: _____

Time: _____

Rank: _____



SQL DATABASE FUNDAMENTALS (345) **STATE 2021**

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 10 points each) _____ (500 points)

TOTAL POINTS _____ (500 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

MULTIPLE CHOICE

Identify the letter of the choice that best completes the statement or answers the question. Mark A if the statement is true. Mark B if the statement is false.

1. The main purpose of constraints is to _____.
 - A. limit what data is allowed in a table
 - B. define how other tables reference a given table
 - C. restrict which queries can be used to modify a table
 - D. none of the above
2. Constraints can be used on columns and tables.
 - A. True
 - B. False
3. If records in a table *cannot* have a NULL value for a column, then that column should have what type of constraint?
 - A. UNNULLABLE
 - B. NOT NULL
 - C. NULLABLE
 - D. UNIQUE
4. Suppose the table Products has the column Price which holds amount of money each product costs. A product must have a price. What constraint is necessary for this column?
 - A. NOT NULL
 - B. UNIQUE
 - C. DEFAULT
 - D. None of the above
5. The INDEX constraint creates an index of all the columns in a table.
 - A. True
 - B. False
6. Which constraint is used to ensure all values of a column meet a certain condition?
 - A. CHECK
 - B. WHERE
 - C. ON
 - D. VERIFY

7. In SQL Server, what type of join is performed by the following query: `SELECT * FROM Student AS s JOIN Enrollment AS e ON s.Id = e.StudentId`
- A. Left Outer Join
 - B. Right Outer Join
 - C. Inner Join
 - D. Full Outer Join
8. Suppose we have two tables, Student and Enrollment. Which type of join would return all the Student records and the matched Enrollment records, assuming Student is the left table and Enrollment is the right table?
- A. Right Outer Join
 - B. Full Outer Join
 - C. Left Outer Join
 - D. Inner Join
9. The WITH clause can be used to join tables based on a condition.
- A. True
 - B. False
10. UNION ALL is different from UNION because _____.
- A. UNION ALL selects duplicate values
 - B. UNION only selects duplicate values
 - C. UNION ALL selects all the records in both tables
 - D. UNION ALL can be used with any number of tables

11. Look at the following SQL query:

```
SELECT p.Name  
FROM Person AS p
```

UNION

```
SELECT e.ID, e.EmployeeName  
FROM Employee AS e;
```

What is wrong with this query?

- A. Both SELECT statements do not select the same number of columns.
 - B. The SELECT statements do not query from the same table.
 - C. UNION ALL should be used instead of UNION.
 - D. None of the above.
12. Each SELECT statement in a UNION must have the same number of columns.
- A. True
 - B. False

13. Each SELECT statement in a UNION can have different types of columns.
- A. True
 - B. False
14. Which command is used to combine the results of two or more SELECT statements?
- A. JOIN
 - B. UNION
 - C. AND
 - D. MINUS
15. Which command is used to combine the distinct results of two or more SELECT statements?
- A. SELECT DISTINCT
 - B. JOIN DISTINCT
 - C. UNION ALL
 - D. SELECT ALL
16. Microsoft SQL Server uses the _____ dialect of SQL, while Oracle uses _____.
- A. PL/SQL; T-SQL
 - B. PL/SQL; PL/pgSQL
 - C. PL/pgSQL; PL/SQL
 - D. T-SQL; PL/SQL
17. What is a schema in Microsoft SQL Server?
- A. A collection of database objects
 - B. The definition of a table in a database
 - C. The definition of a database in a server
 - D. None of the above
18. In SQL Server, a _____ is a set of tables that contains a database's definition.
- A. Schema
 - B. Data dictionary
 - C. Index
 - D. View
19. A foreign key constraint can be added to a table after the table is created.
- A. True
 - B. False

20. Which query correctly adds a foreign key to an existing table?
- A. ALTER TABLE Person ADD FOREIGN KEY (ID) REFERENCES Person(ID);
 - B. UPDATE TABLE Person ADD FOREIGN KEY (ID) REFERENCES Person(ID);
 - C. ALTER TABLE Person FOREIGN KEY (ID) REFERENCES Person(ID);
 - D. UPDATE TABLE Person FOREIGN KEY (ID) REFERENCES Person(ID);
21. Which of the following data type categories does *not* exist in all SQL databases?
- A. Numeric
 - B. Spatial
 - C. Character/String
 - D. Date/Time
22. What is the difference between the int and bigint types in Transact-SQL?
- A. Bigint can hold decimals, int can only hold integers
 - B. Bigint takes up 2 bytes of space, int takes up 4 bytes
 - C. Bigint takes up 8 bytes of space, int takes up 2 bytes
 - D. Bigint takes up 8 bytes of space, int takes up 4 bytes
23. Which of the following Transact-SQL data types would be best suited for storing a person's birthday (storing only the day when the person was born)?
- A. Datetime
 - B. Date
 - C. Time
 - D. Datetimeoffset
24. Which of the following Transact-SQL data types would be best suited for storing the price of an item?
- A. Money
 - B. Decimal
 - C. Numeric
 - D. Float
25. All SQL databases have a data type for storing dates and times.
- A. True
 - B. False
26. Preparing SQL code that can be saved to be executed in the future requires creating a(n) _____.
- A. Index
 - B. Stored Procedure
 - C. View
 - D. Parameterized Query

27. Stored procedures cannot take user defined parameters, a parameterized query must be used instead.
- A. True
 - B. False
28. Stored procedures *cannot* _____.
- A. take parameters
 - B. call other stored procedures
 - C. return a status value to a calling procedure
 - D. all the above can be done with stored procedures
29. Which of the following is a disadvantage of stored procedures?
- A. Stored procedures cannot be easily ported to another RDBMS
 - B. Stored procedures allow for SQL injection from user input
 - C. Stored procedures are not efficient for processing large queries
 - D. None of the above
30. Which command is used to run a stored procedure in SQL Server?
- A. RUN
 - B. START
 - C. EXEC
 - D. STORE
31. Which aggregate function is used to get the largest value of the selected column?
- A. MAX
 - B. MIN
 - C. MAXIMUM
 - D. COUNT
32. Which aggregate function is used to get the number of rows returned by a SELECT statement?
- A. AVG
 - B. NUMBER
 - C. SUM
 - D. COUNT
33. What is the difference between aggregate and scalar functions?
- A. Aggregate functions take many values while scalar functions take one value.
 - B. Aggregate functions return many values while scalar functions return one value.
 - C. Aggregate functions return one value while scalar functions return many values.
 - D. Aggregate functions take one value while scalar functions take many values.

34. The SQL function ABS takes a number and returns the absolute value of that number. Therefore, the ABS function is a(n) _____.
A. Aggregate function
B. String function
C. Scalar function
D. None of the above
35. Scalar functions take one value and return at least one value.
A. True
B. False
36. In relational databases, all transaction must follow _____ principles.
A. CRUD
B. ACID
C. RDBMS
D. BLOB
37. Which of the following statements regarding database transactions is *false*?
A. Transactions provide isolation between programs.
B. Transactions are made up one operation.
C. Transactions represent a single unit of work in a database.
D. Transactions can only modify data in allowed ways.
38. Transactions in relational databases are said to be _____, since they make a single unit of work.
A. Elemental
B. Concrete
C. Static
D. Atomic
39. Which T-SQL command is used to clear all work done by a transaction?
A. ROLLBACK WORK
B. STASH WORK
C. HIDE WORK
D. ERASE WORK
40. Which aggregate function is used to get the average value of a numeric column?
A. NUMBER
B. AVG
C. SUM
D. COUNT

41. Below is a SQL query for the Person and Student tables:

```
SELECT p.Name, s.Grade  
FROM Person AS p  
INNER JOIN Student AS s ON p.Id = s.StudentId;
```

Which of the following best describes the query's results?

- A. The student with the highest grade
- B. The name of every student with a grade greater than 3.0
- C. The name and grade of every person
- D. The name and grade of every person that is a student

42. Which of the following SQL statements could be added to the end of the above query to sort the results by the person's name?

- A. ORDER BY p.Name
- B. GROUP BY s.Grade
- C. ORDER BY s.Grade
- D. GROUP BY p.Name

43. Below is a SQL query for the Student tables:

```
SELECT s.Grade, COUNT(s.Grade)  
FROM Student AS s  
GROUP BY s.Grade;
```

Which of the following best describes the query's results?

- A. The total number of students
- B. The names of all students, grouped by grade
- C. The number of students with each grade
- D. The number of students with the highest grade

44. By default, the ORDER BY keyword sorts results in ascending order.

- A. True
- B. False

45. Below is a SQL query for the Student tables:

```
DELETE FROM Student AS s WHERE s.HasGraduated = 1;
```

What would the query do if the WHERE clause is removed?

- A. Delete all students who have graduated.
- B. Delete all students from the table.
- C. Delete the student table.
- D. None of the above.

46. Which SQL operator returns the results of multiple SELECT statements, where each record is present in all the queried tables?
- A. INTERSECT
 - B. UNION
 - C. UNION ALL
 - D. None of the above
47. Unlike the UNION operator, every SELECT statement used with INTERSECT must have the same number and type of columns selected.
- A. True
 - B. False
48. Data that is in 3NF is said to be more _____ compared to data in 0NF.
- A. clean
 - B. secure
 - C. redundant
 - D. normalized
49. Every student in a high school has a homeroom class, and each homeroom class can contain multiple students. What type of relationship is being described?
- A. One-to-one
 - B. One-to-many
 - C. Many-to-many
 - D. Many-to-none
50. Every student in a high school has a locker, and each locker belongs to only a single student. What type of relationship is being described?
- A. Many-to-many
 - B. Many-to-none
 - C. One-to-one
 - D. One-to-many